

Human-Signature Psychology Relation: A Time-Series Analysis

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ABSTRACT

In this paper, a novel approach is proposed to analyze human signature in time series to predict the author's psychology in what he/she has made his/her signature as the crime rates are increasing day-by-day and need to take special care. The spatial 2D information is transformed to a single dimension time series data by the proposed new method which identifies the minimum and maximum pen change in signature with time interval. This change is then used for statistical feature extraction by applying discrete cosine transform (DCT). The feature is transform invariant and so classification for signer's psychology with k -NN classifier becomes easy. The proposed system is also used for author verification and it gives very good result compared with the other existing methods.

Categories and Subject Descriptors

I.4 [Image Processing and Computer Vision]: Digitization and Image Capture, Segmentation; I.5 [Pattern Recognition]: Clustering, Application, Implementation.

General Terms

Algorithms, Performance, Reliability, and Experimentation.

Keywords

Biometric; Cognitive Computing; Human Psychology; Image Processing; Time Series Signature Analysis;

1. INTRODUCTION

In this ubiquitous world of ubiquitous devices, humans are becoming dependent on such devices for day-to-day living that leads to a ubiquitous life style. These technologies are making the life more simple and comfortable. On the other side, the technologies are misused such as mobile phone with camera may be used in cybercrimes. Thus, if we consider human being as a device then the internal mental status are functionality of that device and can be controlled by manipulating these mental statuses. Modeling human behaviors we can study the mental situation of a human being he/she is and know its interstate transition probabilities. The model may be hierarchically arranged

to identify the short-term and long-term behavior of a human; for example in case of signature, the long-term behavior may be focusing on the signature while the short-term would be maintaining the signature alignment and looking for next job.

These models may be used to improve the human-computer interaction systems. Understanding human behavior, machines/computers can anticipate the human's psychology and adopt it for the better needs of human [1]. To compute the power of the human mind has been a matter of interest from past centuries by the common man, philosopher, psychologist, and educationist [2]. Since the human's internal mental states are not directly visualized and analyzed, an indirect approach must be used. Casual observation and introspection are not sufficient to provide a scientific proof of mental state existent but can be estimated approximately.

Handwritten is one of the indirect ways to map the internal mental state of a human being while he is writing. Hence, handwriting is a brain writing of the writer and predicting the personality using handwritten text is very old. The psychologist studies the frozen graphic structures of a human which is generated in the writer's mind and predict the mental state. If this frozen graph is placed on a piece of paper that is what handwriting is [3]. Similarly, signature is also a frozen graph of a writer while writing in his/her conscious state of mind. The best point in handwritten text is that it is quite easy to identify the psychological state by reading the contents and analyzing the pattern of running writing. But this is not the case with signature. Peoples are more conscious while he/she is giving his/her signature. Simanova et al. [4] have studied the brain relationship aiming to decode semantic categories while participants were speaking, writing, listening to natural sounds and viewing photographs. The snapshot of brain instant of human while writing is shown in Figure 1 where left hemisphere survived the threshold derived by authors.

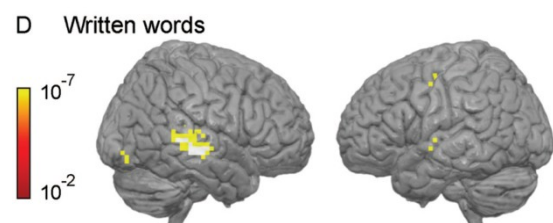


Figure 1. Results for written word-based classification [4].

These studies converges somewhere to have a brain-computer interaction (BCI) [2, 4]. In BCI brain signals are used to operate computers and machine like wheelchair in case of paralyzed patients. The cognitive computing (CC) is involved to model more accurately the 'cognitive' functions of human brain/ mind,

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ICTCS '14, November 14 - 16 2014, Udaipur, Rajasthan, India
Copyright 2014 ACM 978-1-4503-3216-3/14/11...\$15.00
<http://dx.doi.org/10.1145/2677855.2677857>

sensors, reasons, and response to stimulus. Here, in this paper the signature is given main attention to predict the internal mental status in what state he/she signed, forcefully or normal.

In [3], Prasad et al. proposed an interesting tool to predict the human personality based on their handwritten texts. They involve support vector machine (SVM) for classifying the statistical information extracted from writing. The term personality has been variously defined by G. W. Allport [5], i.e. what sort of a person is?, what is he like?, what his/her emotional and social qualities, his/her drives, sentiments and interest. Siddiqi et al. [6] used artificial neural networks (ANN) and support vector machine (SVM) to classify local and global features like slant, curvature, texture and legibility of characters to identify as male or female writer. Whereas in [7] Venkatesh et al. proposed an intelligent system to analyze authors signature to describe their temperaments while he/she singing. This analysis can be used in forensics, in case of forge signature or forcefully made to sign. Can also be used in human resourcing and understanding persons they are dealing with day-to-day life.

The graphologist predicts the personality on the basics of their brain writing and the same is the automated by many researchers since few decades. Thus, in this paper, signature image is taken to analyze in time-series domain to detect in what emotional temperament the person had signed. The rest of the paper is distributes as: second section describes the proposed methodology, third covers the experimental details and results and in the last conclusion is highlighted.

2. PROPOSED METHODOLOGY

There are various works ongoing in field of handwritten analysis for different applications. In personality prediction of human beings based on their handwriting, authors have used various combinations of baseline slant [3, 8, 9], pen pressure [9], character spacing [3], ect. Mogharreban et al. [8] used fuzzy logic to analyze the handwriting of the author whereas in [9], Champa et al. applied ANN to predict the behavior of a human using their handwriting. In this, authors have used baseline, pen pressure and pattern from letter 'r' as input to ANN to output the personality traits of an individual. The literatures are limited to handwritten texts and were analyzed in spatial domain but our case it is extended to signature of individuals.

Here, in this paper, the spatial image information is analyzed in time-domain using time series analysis methods. That is, two dimensional information is transformed to 1D for rotation and translation invariant in formation. The complete flow diagram of the proposed system is shown in Figure 2 and described in next subsections.

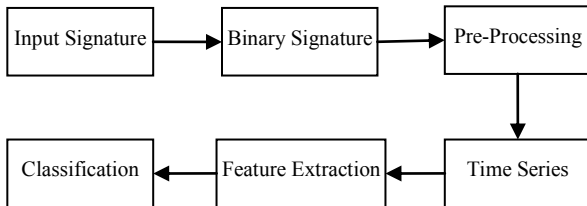


Figure 2. Complete flow diagram of proposed system.

2.1 Image Acquisition and Digitization

The user is asked to give his/her signature on a plain white sheet of paper which is scanned to form a digital image of matrix values. It's not compulsion to use white sheet, different plain

color sheets may also be used since we are filtering all the color information and only retaining the binary information. The sheet color should not match with the ink color used for signature and/or must be easily distinguished. Let say the signed image is S_{rgb} with a resolution $(x * y)$ where x and y depends on the device used in scanning and rgb is the color space model.

2.2 Binary Segmentation

S_{rgb} is transformed to a binary image where the signature is in high intensity and background is in low intensity. This is made possible by passing the image S_{rgb} through a low pass filter, say f_{low} . This filter uses a threshold value, th_α to separate the foreground with the background, i.e. $S_{rgb} = \{S_{rgb_fore}, S_{rgb_back}\}$. Therefore, colored image signature S_{rgb} is converted to binary image S_{bw} using Equation 1. Figure 3 shows the input sign image and binary sign image for more clearance.

$$S_{rgb} \xrightarrow{th_\alpha} S_{bw} \quad \dots (1)$$

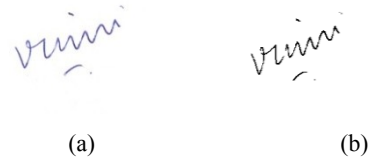


Figure 3. (a) Original input sign image, S_{rgb} of 'Vinni', and (b) binary sign image, S_{bw} .

2.3 Pre-Processing

This binary image S_{bw} is pre-processed which includes normalization and image cropping. In normalization, S_{bw} is inverted, that is intensity value '1' is set to '0' and vice-versa, see in Figure 4. That is, S_{bw} is inverted to form S'_{bw} which is cropped for the actual signature and remove the unwanted background. To obtain this, first top row wise, left column wise, bottom negative row wise and right negative column wise white pixels are located and then forming a rectangle the image is cropped. The size of the rectangle can be derived as $((top, left), (bottom - top, right - left))$. The cropped image, let say it as S'_{crop} , is with the resolution $(x' * y')$ which is always less then $(x * y)$, i.e. $x' < x$ and $y' < y$.



Figure 4. Inverted and cropped binary image, S'_{crop} .

Since every user have different signature with different length there must be a standard format in which the signature is analyzed. Otherwise, the time series transformation will vary. Thus, S'_{crop} is normalized to a square matrix signature, S'_{norm} . The normalized square matrix image is shown in Figure 5 which is further transformed to a time series data.

$$S'_{bw} \xrightarrow{crop} S'_{crop} \xrightarrow{norm} S'_{norm} \quad \dots (2)$$



Figure 5. A normalized input for time series transform.

The resolution of this normalized signature is $(u * u)$ where u is defined by the programmer, which in our case considered as 100, i.e. $u = 100$.

2.4 Time Series Transform

First let understand the concept behind using time series for signature analysis in this paper. The time series is a set of points that is measured in sequence with time change at a uniform time interval. Whereas time series analysis is the process of extracting the meaningful statistical information from the time series data. This analysis may help in prediction and forecasting the futuristic values such as weather forecasting. Since time series has natural temporal ordering, it makes time series analysis very different from normal analysis. Similarly, signature is also a time event where in the pen movement is changed according to the time, T . This time interval measure is the time series data for a signature image, S'_{norm} .

The S'_{norm} is transformed to a time series vector, $\mathcal{V}_S$ by simply summing all the white pixels in a column in such a fashion that each column in an image S'_{norm} is the time instant for time series vector $\mathcal{V}_S$. The equation can be derived as:

$$\mathcal{V}_S(j) = \sum_{i,j=1}^u S'_{norm}(i,j) \quad \dots (3)$$

where (i,j) are the rows and columns of S'_{norm} with maximum value u , and $\mathcal{V}_S(j)$ is the total sum of j column at j time interval. The same is shown in Figure 6 explaining the concept in more detail.

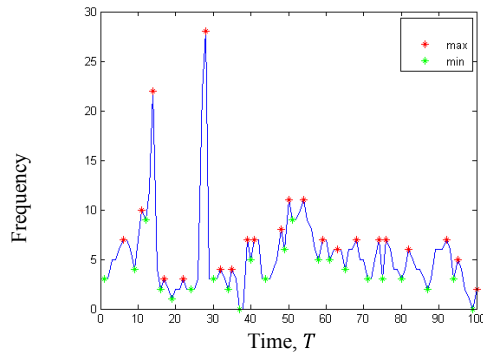


Figure 6. The time series transformation from 2D image to 1D.

Here, in Figure 6, the graph in blue is the S'_{norm} in 1D where, the green stars shows the minimum pen movement while he/she was signing and the red star show that at that instant of time the author was more conscious and gives important by pressing the pen. The x-axis shows the time instants of S'_{norm} and y-axis details the total frequency of white pixels, again in S'_{norm} . The green stars are shows where the letter ends in a signature which is very clear in Figure 7. In this, a red line tries to show the rough separation space of each characters in S'_{norm} .

From this $\mathcal{V}_S$, two separate vectors are formed such that one is a set of maximum points in $\mathcal{V}_S$ and another is the minimum points

set, i.e., $\mathcal{V}_{S_{min}}, \mathcal{V}_{S_{max}} \in \mathcal{V}_S$. Now this will be used as input to discrete cosine transform (DCT) to extract the features, explained in next subheading.

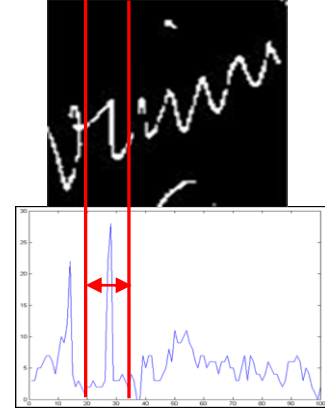


Figure 7. The rough character space in time series for the signature, S'_{norm} .

2.5 DCT Feature Extraction

Since discrete cosine transform is very well known transform and due to the limitation of this paper, we will not cover much details of it and can be referred to [10] for more study. In short, DCT is a finite sequence of data reference derived by sum of cosine functions oscillating at different frequencies. It uses only real numbers with even symmetry, unlike to the discrete Fourier transform (DFT). Suppose, the length of $\mathcal{V}_S$ is N then there are N real numbers, $\{v_{S1}, v_{S2}, \dots, v_{S(N-1)}\}$, which is to be transformed to N real numbers, $\{\mathcal{V}_{S1}, \mathcal{V}_{S2}, \dots, \mathcal{V}_{S(N-1)}\}$ using equation 4 given below.

$$\mathcal{V}_{Sk} = \sum_{n=0}^{N-1} v_{Sn} \cos\left[\frac{\pi}{N} \left(n + \frac{1}{2}\right)k\right] \quad \dots (4)$$

Here, $k = \{0, 1, \dots, (N-1)\}$. This $\mathcal{V}_{Sk}$ is derived for both min and max of $\mathcal{V}_S$ for S'_{norm} . Appending both the DCT vectors in a sequence, that is min followed by max, a feature vector, F_S is formed. The graph for same is shown in Figure 8. Now this F_S is used in classifier for classification.

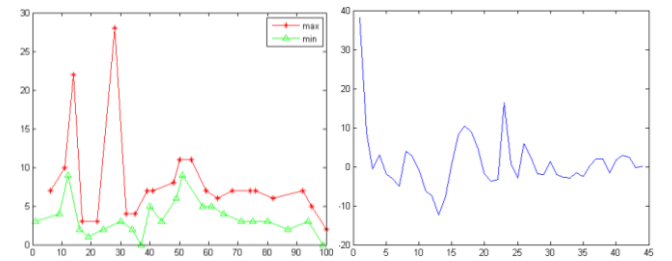


Figure 8. (a) Min-Max graph for $\mathcal{V}_S$, and (b) is DCT feature vector graph.

2.6 Classification

The feature vectors are classified by using a simple popular non-parametric k - Nearest Neighbors k -NN classifier [11]. k -NN is used as a classifier and can also be used for regression. The signature is identified by a majority vote of its neighbors with k nearest neighbors and so it is an instance-based learning. k -NN is termed as lazy classifier as all the computation differed until classification. Therefore, a signature is first classified on basis of author's identity then the features are used for cross verification to

find the psychology of the same. The next section talks about the detailed experimental results and analysis.

3. EXPERIEMNTS AND RESULTS

The proposed system is tested on homemade dataset comprising of 5 different authors with their different mood in different conditions. The psychological moods chosen here for experiment are: normal, hurry, and angry. Each author has been asked to sign in these different moods on different conditions in different days. Total one week it took to collect these samples with 10 samples of each mood by each authors, that is, total 150 signature samples. The method is also explored on different dataset too for signature verification.

The experiment was carried on simple Core2Duo processor with Windows environment in Matlab R2012b. The performance of proposed system is shown in Table I, describing the confusion matrix of different psychological moods. The analysis says that a person in angry mood or if he/she is forced to sign the time series graph is more pressured and the average frequency increases compared to the other moods. Thus, it is easily identified with a very small error rate, shown in Table I. Whereas, in case of hurry, the change is seen at the end of the signature, in maximum time. But it may be the actual signature pattern, so error rate is comparatively more in this case.

In Figure 9, author have signed in two different mood where the second signature is in hurry mood, as seen the tail of the signature is almost messed up compared to the first normal mood. And the signature underline is very small and comparatively far from the actual signature.

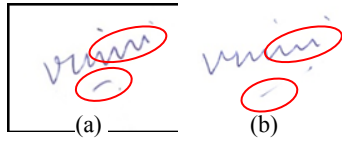


Figure 9. Psychology: (a) normal signature, and (b) hurry mood.

TABLE I. Psychology confusion matrix.

	Normal	Hurry	Angry
Normal	7	2	1
Hurry	3	5	2
Angry	1	3	6

There is another comparison in which the signatures are used to verify the correct author. This approach may be used for authentication and in security systems. The Table II shows a comparison between different datasets and methods and found that the accuracy value is best in our case. This can be even extended to recognize the basic characters of different languages and used in language identification, as proposed by Prased et al. [13]. Time-series can be used to reduce the computation cost of language identification to such an extent and outperform with Curvelet [14], complex algorithm.

TABLE II. Comparison of different methodology with different datasets.

	Datasets	
	SVC2004	Homemade
Krzysztof et al. [12]	88.42	88.96
Proposed	95.21	100

Over this, the computational complexity of the proposed system is comparatively very less and acceptable with such a high rate of accuracy.

4. CONCLUSION

The methodology proposed in this paper precisely analyzes the signature time series to discriminate the psychology of an author while he/she is placing his/her signature on any document(s). A novel approach of time series with min-max DCT feature is used for signature verification and human psychology prediction. The algorithms used are very simple and powerful to distinguish the difference between the intra class sub-classification, shown in Figure 10, understanding the cognitive concept while signing. k -NN is used for classification and comparatively resulted better than others. The overall accuracy rate by our proposed system in signature verification is good but same is not true for psychology prediction. This may be due to poor dataset collected and thus can be improved if dataset is improved and verified by EEG output. Therefore, leaving a future gap for research to continue.

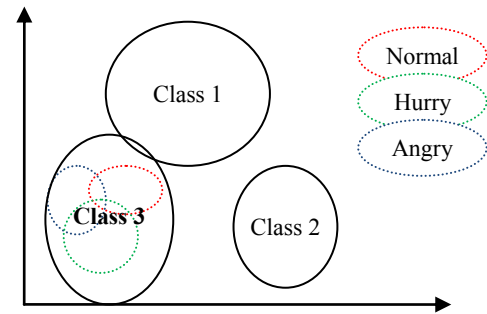


Figure 10. Classes and intra classes classification.

5. REFERENCES

- [1] A. Pentland, and L. Andrew. Modeling and prediction of human behavior. *Neural computation*. 11, 1 (1999), 229-242.
- [2] P. E. Vernon, The Structure of Human Abilities (Psychology Revivals). Routledge, (2014).
- [3] S. Prasad, V. K. Singh, and A. Sapre. Handwriting Analysis based on Segmentation Method for Prediction of Human Personality using Support Vector Machine. *International Journal of Computer Application* (0975-8887). 12, (2010), 8.
- [4] I. Simanova, P. Hagoort, R. Oostenveld, and M. A. van Gerven. Modality-independent decoding of semantic information from the human brain. *Cerebral Cortex*, 24, 2 (2014), 426-434.
- [5] G. W. Allport. Personality: A psychological interpretation. (1937).
- [6] I. Siddiqi, C. Djeddi, A. Raza, and L. Souici-meslati. Automatic analysis of handwriting for gender classification. *Pattern Analysis and Applications*. (2014), 1-13.
- [7] A. Venkatesh, A. Bhanpurawala, S. Chowdhury, D. R. Kalbande. Demonstration of Signature Analysis Using Intelligent Systems. *Advances in Computing, Communication, and Control*. Springer Berlin Heidelberg , (2013), 199-206.
- [8] N. Mogharreban, S. Rahimi, and M. Sabharwal. A combined crisp and fuzzy approach for handwriting analysis. *IEEE Annual Meeting of the Fuzzy Information Processing, NAFIPS 2004*. 1, (2004).
- [9] H. N. Champa, K. R. Ananda Kumar. Artificial Neural Network for Human Behavior Prediction through Handwriting Analysis. *International Journal of Computer Applications, IJCA*. 2, 2, (2010), 36-41.

- [10] N. Ahmed, T. Natarajan, and K. R. Rao. "Discrete cosine transform." *Computers, IEEE Transactions on* 100.1 (1974): 90-93.
- [11] Tomašev, Nenad, et al. "Hubness-based fuzzy measures for high-dimensional k-nearest neighbor classification." *International Journal of Machine Learning and Cybernetics* 5.3 (2014): 445-458.
- [12] C. Krzysztof, M. Zalasinski, and L. Rutkowski. "New method for the on-line signature verification based on horizontal partitioning." *Pattern Recognition* 47.8 (2014): 2652-2661.
- [13] S. Prasad, G. K. Verma, B. K. Singh, and P. Kumar, "Basic Handwritten Character Recognition From Multi-Lingual Image Dataset Using Multi-Resolution And Multi-Directional Transform," *Int. J. of Wavelets, Multiresolution and Information Processing*, 10(05), (2012).
- [14] G. K. Verma, S. Prasad, and P. Kumar, "Handwritten Hindi Character Recognition Using Curvelet Transform," In *Information Systems for Indian Languages*, Springer Berlin Heidelberg, (2011), 224-227.